

Gauranga Kumar Baishya

Chennai Mathematical Institute (CMI)

Whitefield, Bangalore, Karnataka, India

✉ gaurangakumarbaishya@gmail.com • 🌐 <https://gaurangakrb.github.io/>

🌐 <https://www.linkedin.com/in/gauranga-kumar-baishya-4a12731ba/>



Education

Program	Institution/Board	%/CGPA	Year
M.Sc. (Data Science)	Chennai Mathematical Institute (CMI) Chennai, Tamil Nadu	8.50/10	2023-25
B.Sc.(Mathematics)	Tezpur Central University (TU) Napaam, Tezpur, Assam	80.20%	2019-22
Higher Secondary (Science)	Shrimanta Shankar Academy (SSA) Dispur, Guwahati, Assam	87.60%	2016-18
High School	Shrimanta Shankar Academy (SSA) Dispur, Guwahati, Assam	10/10	2015-16

Key Projects & Experience

- 1. Deconvolution Analysis of Human adipose tissue** May 2024 - Mar 2026
Guide: Prof. Yu-Hua Tseng Harvard Medical School, Brookline, MA, USA
 - ML-based deconvolution pipeline for bulk RNA-seq of deep & superficial cervical adipose tissue, modeling cell-type composition and expression signatures to resolve depot-specific transcriptional variance & differential pathways. [†]
- 2. Mechanistic Interpretability of Foundation Models** May 2025 - Feb 2026
Guide: Prof. Amitava Das Birla Institute of Technology (BITS) Pilani, Goa
 - Developed differential geometry-based framework for transformers that, across layers, measures (i) *curvature and path length* of the model's predictive distribution via Fisher–Rao on logits, (ii) *gauge-invariant parallel transport of hidden states*, and (iii) attention as a *depth-indexed transport operator*, producing task-aligned gradient fields that localize where and how the network reorganizes *its beliefs*. [Project page](#)
- 3. Multimodal Single Cell Analysis** Aug 2022 - Apr 2023
Guide: Prof Sandeep Choubey & Prof. Dhiraj Hazra Institute of Mathematical Sciences (IMSc.), Chennai
 - Modeled cross-omic dependencies in single-cell hematopoietic differentiation by applying *Gaussian Processes* and *hierarchical Bayesian inference* with *MCMC sampling* to learn nonlinear covariation structures among DNA, RNA, and protein modalities across temporal developmental trajectories. [Github Repo](#)

Industrial Training

- 1. Lucid Systematic Literature Review (SLR) engine** Aug 2025 - present
Dr. Amrita Ostawal, CEO Arete Access
 - Developing an AI-driven SLR screening model that uses domain-tuned transformers for inclusion/exclusion, RAG over guidelines, & structured information extractors to generate audit-ready, regulatory-grade evidence. [†]
- 2. Modeling of Cadmium Eco-toxicity** Sep 2025 - Nov 2025
Dr. Sunil Nagpal & Mrs. Mitali Merchant Tata Research, Development & Design Centre (TRDDC)
 - Modeling of cadmium ecotoxicity using probabilistic graphical models to identify and predict toxicological pathways and environmental risk patterns; won 1st prize in *Research challenge project*. [†]
- 3. Classification of TILs in OSC carcinoma histopathology** Nov 2025 - Mar 2026
Saipradeep V.G. Tata Consultancy Services - Research, Bangalore
 - Developed ML models for automating TIL quantification for OSCC and a high-performance vector embedding pipeline to discover 'similar' Whole Slide Images (WSIs) and patches in real-time, using Qdrant, FastAPI, and Streamlit. [†]

Key Course Projects

- 1. Course: Applied Machine Learning** Jan-Apr 2025
Instructor: Dr. Raghav Kulkarni, Principal Data Scientist, Success in Cloud Inc., USA CMI

- Empirical benchmark of *classical ML* vs *modern DL* (including custom ANN and BERT/DistilBERT) for Goodreads review-popularity prediction showing boosting can rival transformer recall under severe class imbalance. [Github Repo](#)

2. Course: Guided project/Causal Inference & Modelling

Jan-Apr 2025

Instructor: Prof. M. R. Srinivasan, CMI

CMI

- Causal inference framework integrating *potential outcomes framework*, *DAG-based structural equations modeling*, and *DoubleML orthogonalization* to estimate race-conditioned effects in COVID-19 telemedicine adoption using inverse propensity weighting and counterfactual simulation. [Github Repo](#)

3. Course: Natural Language Processing

Aug-Nov 2024

Instructor: Prof. Ramaseshan Ramachandran, CMI

CMI

- Integrated computational linguistics & representation learning framework combining statistical text analysis, probabilistic semantics, and deep sequence modeling - from *Zipf-Heaps* corpus characterization to *COALS* and skip-gram embeddings through RNN/LSTM-driven abstract generation on $\approx 50,000$ COVID-19 scientific articles. [Github Repo](#)

4. Linear Algebra & its applications

Jan-Apr 2024

Instructor: Prof. Priyavrat Deshpande, faculty, CMI

CMI

- Implemented graph-theoretic image segmentation using spectral clustering; formulating *normalized cut* via Laplacian eigen decomposition & integrating SLIC superpixels with RAG-based recursive partitioning to achieve perceptual grouping; based on the breakthrough paper in 2000 by *Shi & Malik*. [Github Repo](#)

Online Courses

- DeepLearning.AI: *Tensorflow Developer Specialization*, *DeepLearning Specialization*, *Machine Learning Specialization*
- Coursera: *Probabilistic Graphical Models Specialization*, *Introduction to Genomic Data Science*
- Others: *Deepmind x UCL Introduction to RL by David Silver*, *Introduction to Large Language Models (NPTEL)*, *Introduction to Synthetic Biology (IIT Madras)*, *Reinforcement Learning (NPTEL)*

Course Work

Key Courses

Aug 2019-Apr 2022

(Core)

Tezpur University

- Real Analysis - I&II, Calculus - I&II, Introductory ODE&PDE, Linear Algebra - I, Physics - I,II,III, Introductory Statistics and Probability, Introduction to Optimisation, Numerical Methods and Boolean Algebra, Scientific Computing

Key Courses

Aug 2023-Apr 2025

(Core and electives)

CMI

- Programming & Data Structures with Python, RDBMS & SQL, Linear Algebra & its Applications, Regression & Classification, Algorithm Design Techniques, Advanced Machine Learning, NLP, Multivariate Statistics, Applied Machine Learning

Technical Skills

- OS: Windows, MAC, Linux
- Tools: LaTeX, Anaconda, Adobe, Git, MS Office
- Programming Language: C, R, Matlab, Python
- Database System: MySQL

Positions of Responsibility, Volunteering & Social Work

- Course Instructor:** Mathematics for Machine Learning at *Upmenta* (2025 - present).
- Managing Director** of Non-Profit Organisation *Sankhya* (2024 - 2025).
- Lead AI/ML Strategy** at *Arete Access*. (2024 - present).
- Academic mentor** at *Topmate.io* **top 1%** (2024 - present).
- Placement Co-ordinator** for *M.Sc. Data Science Course*, *Chennai Mathematical Institute (CMI)* (2023-24).
- Course instructor:** Mathematics, Statistics & CS for *Cheenta School of Statistics & Data Science, India* (2023-25).
- Event Co-ordinator** at *Polymath Jr. REU, US* (Jun - Aug 2021).

Workshops and Conferences Attended

- Attended *India AI Impact Summit 2026*
- Attended *Explorations in Statistics, Probability, Learning and Optimization Research (exSPLOre) - 2026* conference (2025) funded by *Google Deepmind & Ashoka University*.

- Attended [Data Science, Probabilistic & Optimisation Methods - II](#) conference (2025) funded by *Google Deepmind, ICTS-Bangalore & Ashoka University*. 
- Attended [Geometry of Machine Learning Conference](#) (2025) conference conducted by *CMSA, Harvard University*.
- Attended the Symposium on [Artificial Intelligence and Pharmaceutical Medicine \(2025\)](#), jointly organized by *Pfizer and Indian Institute of Technology (IIT) Madras*. 
- Attended the workshop on [Language & Vision with AI/ML](#) (2025), conducted by *Multimedia Lab, Indian Institute of Technology (IIT) Guwahati*. 
- Attended [Accelerating statistical inference and experimental design with ML](#) (2025) conference, conducted by *Isaac Newton Institute for Mathematical Sciences, University of Cambridge*. 
- Attended the [CCAIM AI and ML Summer School in medicine](#) (2024) & [in healthcare](#) (2023), (with scholarship), conducted by the *University of Cambridge, UK*. , 
- Attended the workshop on [Synthetic Biology](#) (2024), conducted by *Indian Institute of Technology (IIT) Madras*. 
- Participated in the [5 day Generative AI - intensive course](#), conducted by *Google*.
- Attended [International Conference on Security & Privacy](#) (2020), conducted by *National Institute of Technology Jamshedpur*. 
- Attended [International Conference on Number Theory and Discrete Mathematics](#) (2020), conducted by *Ramanujan Mathematical Society*. 

Invited Talks & Workshops

- Conducted a 2-day workshop on the [Fundamentals of Machine Learning](#) for B.Tech - CSE students, conducted by the *IEEE students council, GLA University, Mathura* (2025).
- Appeared for [a podcast](#), a conversation with Mohammed Yousef Shaik (CEO of Approtors Technology Pvt. Limited) about my experiences in Data Science (2025).
- Gave [a guest talk](#), on [Causal Inference](#), organized by the *Google Developer Group, IIT Madras*.
- Gave a [guest public talk](#), on the Life and Work of *John Von Neumann* for high school kids on the occasion of National Mathematics Day (2023), conducted by the *District Commissioner (DC), West Karbi Anglong, Assam*.
- Conducted a [workshop](#) for state math olympiad winners conducted by the *Assam Academy of Mathematics (AAM)*.
- Gave a [talk](#) as a part of the *Polymath REU Jr.* program 2021 listed by the *American Mathematical Society (AMS)*.

Achievements/Awards

- Awarded the prestigious [Khorana Scholars Program Fellowship](#) 2024 ([acceptance rate: 0.2%](#)).
- Secured [North-East India Rank 1](#) (2020) in Madhava Mathematical Competition (equivalent to the William Lowell Putnam Mathematical Competition, USA).
- Qualified [National Talent Search Examination \(NTSE\)](#) Level-1, 2017 ([top 0.5%](#)).
- Secured [All India Rank \(AIR\) 89](#) in the Indian Institute of Technology (IIT) - JAM Mathematics ([top 0.6%](#)).
- Awarded [Dr. Subratananda Dowerah Memorial Gold Medal - 2016](#) by Assam Academy of Mathematics (AAM).
- Awarded [Dr. Durgeswar Das Memorial Award - 2016](#) by Assam Academy of Mathematics (AAM).
- Qualified for [Indian National Mathematical Olympiad \(INMO\)](#) four times in 2014, 15, 16 & 17.
- Selected for the [JNCASR Summer Research Fellowship](#) (2020).
- Selected for the [Mathematics Training and Talent Search \(MTTS\)](#) Programme (2021).
- Secured [Rank 1](#) in State Chemistry Olympiad (2016).
- Awarded [The Best Academic Award](#) by my school, in 2016, for outstanding academic excellence.
- Awarded [School Math Topper Award](#) (2013) in the International Mathematics Olympiad (IMO) conducted by the Science Olympiad Foundation (SOF).

Others

- Hobbies: Solo-Travelling, Reading, Teaching, Sketching
- Languages: English, French (A1), Hindi, Assamese

† Due to confidentiality agreements, I am unable to share the code repositories.